

Homework 2:

Word level bigrams using TF-IDF:

Message 1: $TF = [1/4, 1/4, 1/4, 1/4, 0, 0, 0, 0, 0]$

$IDF = [\ln(3/1), \ln(1/1), \ln(3/2), \ln(3/4), \ln(3), \ln(3), \ln(3), \ln(3), \ln(3)]$

$TF-IDF = [1/4 \ln(3), 0, 1/4 \ln(3/2), 1/4 \ln(3/4), 0, 0, 0, 0, 0]$

Message 2: $TF = [0, 1/5, 0, 0, 1/5, 1/5, 1/5, 1/5, 0]$

$TF-IDF = [0, 0, 0, 0, 1/5 \ln(3), 1/5 \ln(3), 1/5 \ln(3), 1/5 \ln(3), 0]$

Message 3: $TF = [0, 1/4, 1/4, 1/4, 0, 0, 0, 0, 1/4]$

$TF-IDF = [0, 0, 1/4 \ln(3/2), 1/4 \ln(3/2), 0, 0, 0, 0, 1/4 \ln(3)]$

1) TF of Bigrams for Message 2:

$[0, 1/5, 0, 0, 1/5, 1/5, 1/5, 1/5, 0]$

2) IDF of Bigram for Message 2 (IDF same across all 3 messages):

$[1/4 \ln(3), 0, 1/4 \ln(3/2), 1/4 \ln(3/2), 0, 0, 0, 0, 0]$

3) TF-IDF Message 2: $[0, 0, 0, 0, 1/5 \ln(3), 1/5 \ln(3), 1/5 \ln(3), 1/5 \ln(3), 0]$

4) The TF-IDF for "are good" = 0 because this word exists within all 3 documents, and it being 0 makes sense to have no weight on common words and allow unique words to have larger weights.

5) $IDF = \ln\left(\frac{\# \text{ docs}}{\# \text{ docs w/ that word}}\right)$

if both num =, $\ln(1) = 0$. So all words that exists within all 3 docs will have 0 IDF.

• Are good } only this word

2) Support is the # of occurrence of each class in our dataset.

3) No because accuracy will give a false reading when we have high FN or FP's, while F1 score accounts for this. Based on our confusion matrix, FN and FP values aren't too large compared to TN and TP which is good, but not small enough for us to just use accuracy. But in this case, it's not too drastic, so accuracy is ok but F1 is safer.

4) There's 2 methods that I think of:

- Update weights s.t the model will be punished when it misses or predicts a false result esp. if the bad data is the minority.
- Oversample / duplicate the bad samples to be equivalent to the good samples.